1. **Write a Python program to compute the output of a single artificial neuron using the sigmoid activation function. Clearly show weighted sum calculation, bias addition, and final activated output.**

IDLE Shell 3.13.5 - C:/Users/saimo/OneDrive/ドキュメント/Desktop/ML Programs/Model lab 1st Program.py (3.13.5)

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/saimo/OneDrive/ドキュメント/Desktop/lab program 1.py ======
=====
----- Q1: Single Neuron -----
Weighted Sum: 0.5
After Bias: 0.7
Final Output: 0.6682
>>>
```

lab program 1.py - C:/Users/saimo/OneDrive/ドキュメント/Desktop/lab program 1.py (3.13.5)

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```python
import math

x1, x2 = 1, 0
w1, w2 = 0.5, -0.3
bias = 0.2

def sigmoid(z):
    return 1 / (1 + math.exp(-z))

print("----- Q1: Single Neuron -----")

weighted_sum = (w1 * x1) + (w2 * x2)
print("Weighted Sum:", weighted_sum)

z = weighted_sum + bias
print("After Bias:", z)

output = sigmoid(z)
print("Final Output:", round(output,4))
```

2. **Extend the program to perform one iteration of the backpropagation algorithm. Compute the error, update weights using a chosen learning rate, and display updated weight values.**

IDLE Shell 3.13.5 - C:/Users/saimo/OneDrive/ドキュメント/Desktop/ML Programs/lab program 2.py (3.13.5)

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```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>
========== RESTART: C:/Users/saimo/OneDrive/ドキュメント/Desktop/lab program 2.py ======
=====
----- Q2: Backpropagation -----
Error: 0.3318
Updated w1: 0.5074
Updated w2: -0.3
Updated bias: 0.2074
>>
```

lab program 2.py - C:/Users/saimo/OneDrive/ドキュメント/Desktop/lab program 2.py (3.13.5)

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```python
import math

x1, x2 = 1, 0
target = 1

w1, w2 = 0.5, -0.3
bias = 0.2
lr = 0.1

def sigmoid(z):
    return 1 / (1 + math.exp(-z))

print("----- Q2: Backpropagation -----")

z = w1*x1 + w2*x2 + bias
output = sigmoid(z)

error = target - output
print("Error:", round(error,4))

gradient = error * output * (1 - output)

w1 += lr * gradient * x1
w2 += lr * gradient * x2
bias += lr * gradient

print("Updated w1:", round(w1,4))
print("Updated w2:", round(w2,4))
print("Updated bias:", round(bias,4))
```

3. **Implement logistic regression for the same input-output mapping and compare the predicted output with that of the neural network. Explain similarities and differences observed in the results.**

IDLE Shell 3.13.5 - C:/Users/saimo/OneDrive/ドキュメント/Desktop/ML Programs/lab program 3.py (3.13.5)

File Edit Shell Debug Options Window Help

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/saimo/OneDrive/ドキュメント/Desktop/lab program 3.py ======
=====
----- Q3: Logistic Regression -----
Input: [1, 0] Predicted: 0.6682 Actual: 1
Input: [0, 1] Predicted: 0.475 Actual: 1
Input: [1, 1] Predicted: 0.5987 Actual: 1
Input: [0, 0] Predicted: 0.5498 Actual: 0
>>>
```

lab program 3.py - C:/Users/saimo/OneDrive/ドキュメント/Desktop/lab program 3.py (3.13.5)

File Edit Format Run Options Window Help

```
import math

X = [[1,0], [0,1], [1,1], [0,0]]
Y = [1, 1, 1, 0]

w1, w2 = 0.5, -0.3
bias = 0.2

def sigmoid(z):
    return 1 / (1 + math.exp(-z))

print("----- Q3: Logistic Regression -----")

for i in range(len(X)):
    x1, x2 = X[i]

    z = w1*x1 + w2*x2 + bias
    output = sigmoid(z)

    print("Input:", X[i], "Predicted:", round(output,4), "Actual:", Y[i])
```

4. **Run multiple training iterations and analyse convergence behaviour. Explain, using printed outputs, whether the neuron output stabilizes and what this indicates about the learning process.**

IDLE Shell 3.13.5 - C:/Users/saimo/OneDrive/ドキュメント/Desktop/ML Programs/lab program 4.py (3.13.5)

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```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>
========== RESTART: C:/Users/saimo/OneDrive/ドキュメント/Desktop/lab program 4.py ======
=====
----- Q4: Training Iterations -----
Epoch 1 Output: 0.6682
Epoch 2 Output: 0.6714
Epoch 3 Output: 0.6746
Epoch 4 Output: 0.6778
Epoch 5 Output: 0.6808
Epoch 6 Output: 0.6838
Epoch 7 Output: 0.6868
Epoch 8 Output: 0.6897
Epoch 9 Output: 0.6925
Epoch 10 Output: 0.6953

Final Output Stabilized: 0.6953
>>
```

lab program 4.py - C:/Users/saimo/OneDrive/ドキュメント/Desktop/lab program 4.py (3.13.5)

File Edit Format Run Options Window Help

```
import math

x1, x2 = 1, 0
target = 1

w1, w2 = 0.5, -0.3
bias = 0.2
lr = 0.1

def sigmoid(z):
    return 1 / (1 + math.exp(-z))

print("----- Q4: Training Iterations -----")

for epoch in range(10):

    z = w1*x1 + w2*x2 + bias
    output = sigmoid(z)

    error = target - output
    gradient = error * output * (1 - output)

    w1 += lr * gradient * x1
    w2 += lr * gradient * x2
    bias += lr * gradient

    print("Epoch", epoch+1, "Output:", round(output,4))

print("\nFinal Output Stabilized:", round(output,4))
```